

Q5 Ans: A

The effect of atmospheric pressure cancels out at the top and bottom of the drum.

At equilibrium,

weight of drum & content = weight of water displaced

$$= V\rho g$$

$$= 0.80 \times 0.19 \times 1.1 \times 10^3 \times 9.81$$

$$= 1640 \text{ N}$$

$$\text{mass of drum \& content} = \frac{1640}{9.81} = 167 \text{ kg}$$

Q6 Ans: C